THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 4: **Quadratic Equations** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is **not** a quadratic equation? 1

(a) $(x+1)^2 = 2(x-3)$ **(b)** $x^2 - 2x = (-2)(3-x)$ **(c)** $(x+1)^2 = x(x+3)$ **(d)** $(x-3)(2x+1) = x(x+5)$

2. The roots of $2x^2 - 5x + 3 = 0$ are: 1

(a) $1, \frac{3}{2}$ **(b)** $-1, -\frac{3}{2}$ **(c)** $2, 3$ **(d)** $1, -\frac{3}{2}$

3. The discriminant of $2x^2 + 5x - 3 = 0$ is: 1

(a) 49 **(b)** 1 **(c)** −49 **(d)** 31

4. If the discriminant of a quadratic equation is negative, its roots are: 1

(a) real and distinct **(b)** real and equal
(c) not real **(d)** zero

5. A student states: “Every quadratic equation has exactly two real roots.” The correct response is: 1

(a) Correct **(b)** Incorrect; e.g. $x^2 + 1 = 0$ has no real root
(c) Correct for $a > 0$ **(d)** Cannot be determined

6. If one root of $x^2 + px - 6 = 0$ is 2, then p is: 1

(a) 1 **(b)** −1 **(c)** 2 **(d)** 3

7. The values of k for which $4x^2 - 3kx + 1 = 0$ has equal roots are: 1

(a) $\pm\frac{3}{4}$ **(b)** $\pm\frac{4}{3}$ **(c)** ± 2 **(d)** ± 1

8. For $x^2 + x - 6 = 0$, Student A gets roots $2, -3$ and Student B gets $-2, 3$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

9. A quadratic equation whose roots are 2 and -5 is: 1

(a) $x^2 + 3x - 10 = 0$ **(b)** $x^2 - 3x - 10 = 0$ **(c)** $x^2 + 7x + 10 = 0$ **(d)** $x^2 - 3x + 10 = 0$

10. “The sum of a number and its reciprocal is $\frac{10}{3}$” is represented by: 1

(a) $3x^2 - 10x + 3 = 0$ **(b)** $3x^2 + 10x + 3 = 0$ **(c)** $x^2 - 10x + 3 = 0$ **(d)** $3x^2 - 10x - 3 = 0$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve by factorisation: $100x^2 - 20x + 1 = 0$. 2

12. Find the discriminant of $3x^2 - 4\sqrt{3}x + 4 = 0$ and hence the nature of its roots. Find the roots if real. 2

13. Find k such that $kx^2 + 2x + 1 = 0$ has equal roots. For which values of k will it have no real roots? 2

14. A student solved $x^2 - 2x - 15 = 0$ and wrote the roots as 5 and 3. Check whether $x = 3$ is a root and give the correct roots. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve using the quadratic formula: $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$. 3

16. Find two consecutive odd positive integers, the sum of whose squares is 290. 3

17. The diagonal of a rectangular field is 60 m more than the shorter side. If the longer side is 30 m more than the shorter side, find the sides of the field. 3

OR

Find the values of k for which $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots. 3

18. Is it possible to design a rectangular mango grove whose length is twice its breadth and area is 800 m²? If so, find its length and breadth. 3

OR

A student claims that a rectangle with perimeter 28 m and area 50 m² can be made. Justify or refute. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train and verify. 5

OR

In a class test, the sum of Shefali's marks in Mathematics and English is 30. Had she got 2 marks more in Mathematics and 3 marks less in English, the product of her marks would have been 210. Find her marks in the two subjects. 5

20. **Case study:** Rohan's mother is 26 years older than him. The product of their ages 3 years from now will be 360. 5

(i) Form the quadratic equation.

(ii) Find Rohan's present age.

(iii) Find his mother's present age and verify.

OR

A pole has to be erected at a point on the boundary of a circular park of diameter 13 m such that the difference of its distances from two diametrically opposite fixed gates A and B is 7 m. Is it possible? If yes, at what distances from the two gates should it be erected? 5

END OF QUESTION PAPER — Total: 40 Marks